Manufacturing Enemies in Political Contests

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Abstract

I develop a model of electoral competition in an identity-fragmented economy where Majority parties differ in technological competence: one excels in public good provision (the *Dove*), while the other excels in security and identity consolidation (the *Hawk*). Symbolic policies provoke Minority mobilization, and higher conflict widens the Hawk's relative advantage in providing security by amplifying the returns to security competence and alleviating the Majority's collective action problem. With a fixed voter trade-off between security and consumption, the Hawk may optimally choose provocative symbolic policy when the electoral gain from a larger Security Premium outweighs its distortion/backlash costs. A dynamic extension with noisy performance signals, myopic retrospective voting, and heterogeneous voter patience generates regime switching between high-d/low-g and low-d/high-g regimes, and clarifies when the economy converges to a conflict trap versus political cycles.

1 The Model Environment

Consider an economy populated by a measure one continuum of citizens divided into two distinct ethnic or religious groups: a Majority M and a Minority m. The Majority comprises a population share $n_M > 1/2$, while the Minority comprises the remainder $n_m = 1 - n_M$. The Majority group is represented by two competing political parties, Party A (the *Hawk*) and Party B (the *Dove*), who compete for the electoral support of the Majority representative voter.

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The incumbent party $k \in \{A, B\}$ selects a policy vector $\mathbf{q}_k = (d_k, g_k) \in \mathbb{R}_+^2$ subject to their technological constraints. The policy consists of two instruments. First, the symbolic policy d represents the intensity of identity-based regulations, such as cultural bans or dress codes, which serve as a coordination device for identity grievances. Second, economic policy g captures the provision of public policies on public goods provision, market integration, or employment opportunities.

Parties are distinguished by their technological capabilities in delivering these policies. We assume that Party A possesses a comparative advantage in the production of security and coercion, denoted by competence parameter S_A. Conversely, Party B possesses a comparative advantage in economic management but holds inferior security competence S_B. Formally, we posit that $S_A > S_B$, while the cost of providing public goods satisfies $c_A(g) > c_B(g)$ for any equivalent level of g.

1.1 Timing and solution concept

The political interaction is a three-stage game solved by backward induction.

Stage 1: Policy choice. The incumbent party $k \in \{A, B\}$ selects a policy vector (d_k, g_k).

Stage 2: Contest and mobilization. Given the symbolic environment induced by d_k, the Minority mobilizes, and Majority citizens choose enforcement effort given the incumbent's security competence S_k.

Stage 3: Election. The Majority electorate votes between the Hawk and the Dove. In a stationary equilibrium, voters evaluate each party using the equilibrium outcomes implied by backward induction.

The equilibrium concept is subgame perfect equilibrium. In the contest subgame among Majority citizens, I focus on symmetric Nash equilibria.

2 Micro-foundations of Conflict and Collective Action

Conflict in this economy is modeled as a contest over political dominance. Unlike standard contest models where groups act as unitary actors, I explicitly model the collective action problem within each group.

2.1 The Minority's Reaction

Minority mobilization is an endogenous response to the grievance generated by symbolic policy. The representative minority member chooses effort to minimize the loss from symbolic oppression net of effort costs.

Formally, the representative agent chooses effort level e_m to maximize a utility function $V(e_m; d)$. This function balances the psychological gratification of resistance against the convex cost of mobilization:

$$\max_{e_m \geq 0} \quad V(e_m; d) = \underbrace{\phi d^\sigma e_m}_{\text{Expressive Utility}} - \underbrace{\frac{1}{2\kappa} e_m^2}_{\text{Cost of Effort}} \tag{1}$$

The first term captures the marginal benefit of resistance, where $\phi > 0$ represents the group's *sensitivity to grievance* and d^σ captures the intensity of the symbolic shock (with elasticity $\sigma > 0$). The second term represents the cost of mobilization, where $\kappa > 0$ denotes the group's *organizational capacity* (or social capital). A higher κ reduces the marginal cost of effort, reflecting stronger internal coordination networks.

The first-order condition (FOC) for an interior solution implies:

$$\frac{\partial V}{\partial e_m} = \phi d^\sigma - \frac{1}{\kappa} e_m = 0. \tag{2}$$

Solving for e_m, we derive the structural reaction function for aggregate minority mobilization:

$$E_m(d) = \gamma d^\sigma, \tag{3}$$

where $\gamma \equiv \phi\kappa$ is the *structural mobilization potential*. This explicit derivation clarifies that the opposition's strength γ is a composite of two forces: the intensity of their grievance sensitivity (ϕ) and their capacity to organize (κ).

2.2 The Majority's Collective Action Problem

While both economic stability and security are public goods, they differ in their production technology. Economic goods are provided top-down by the state, whereas security (dominance) requires decentralized participation by Majority citizens. This creates a standard free-rider problem: each Majority citizen benefits from dominance, but individually underprovides enforcement effort.

Fix an incumbent party k and suppose the Majority consists of N_M identical citizens

indexed by $i = 1, \ldots, N_M$. Citizen i chooses enforcement effort $e_i \geq 0$ at private cost $\frac{1}{2}e_i^2$. Aggregate Majority effort is $E_M \equiv \sum_{j=1}^{N_M} e_j$. Minority mobilization E_m is determined by Equation 3 following the symbolic policy choice.

Security is produced by a stochastic contest between a state-amplified Majority and an oppositional Minority. Let $S_k > 0$ denote the incumbent's security competence, which acts as a force multiplier on Majority effort. Following standard contest theory, the probability that the Majority prevails is given by the Tullock contest success function:

$$\Pi_k(E_M, E_m) = \frac{S_k E_M}{S_k E_M + E_m}. \tag{4}$$

[1] This formulation captures the model's central asymmetry: holding office grants access to S_k, which scales the effectiveness of decentralized Majority effort, and therefore makes security competence electorally salient when the opposition is mobilized.

Given (S_k, E_m), each Majority citizen i chooses e_i to maximize private utility,

$$\max_{e_i \geq 0} \quad W_i = \underbrace{\Lambda \cdot \Pi_k(E_M, E_m)}_{\text{Public Benefit}} - \underbrace{\frac{1}{2}e_i^2}_{\text{Private Cost}} \tag{5}$$

where $\Lambda > 0$ is the per-capita valuation of security. Taking aggregate effort of other citizens as given, the first-order condition for an interior best response is

$$e_i = \Lambda \frac{\partial \Pi_k}{\partial E_M}(E_M, E_m) = \Lambda \frac{S_k E_m}{(S_k E_M + E_m)^2}. \tag{6}$$

In a symmetric Nash equilibrium, $e_i = e$ for all i, so $E_M = N_M e$ and aggregate Majority mobilization $E_M^*(S_k, E_m)$ satisfies

$$\frac{E_M^*(S_k, E_m)}{N_M} = \Lambda \frac{S_k E_m}{(S_k E_M^*(S_k, E_m) + E_m)^2}. \tag{7}$$

Lemma 1 (Contest equilibrium) *For any $(S_k, E_m) \in (0, \infty) \times [0, \infty)$, the Majority mobilization game admits a unique symmetric Nash equilibrium. If $E_m = 0$, then $E_M^*(S_k, 0) = 0$. If $E_m > 0$, then $E_M^*(S_k, E_m) > 0$ is the unique solution to Equation 7. Define equilibrium security*

$$\Pi_k^*(S_k, E_m) \;\equiv\; \Pi_k\big(E_M^*(S_k, E_m), E_m\big). \tag{8}$$

1. Following the contest theory literature (e.g., Skaperdas (1996); Jia (2008)), one micro-foundation assumes realized "performance" levels $Y_M = \ln(S_k E_M) + \varepsilon_M$ and $Y_m = \ln(E_m) + \varepsilon_m$, where $\varepsilon_M, \varepsilon_m$ are i.i.d. Type I Extreme Value (Gumbel). The probability of Majority dominance is $\Pi_k = \mathbb{P}(Y_M > Y_m)$, which simplifies to Equation 4.

Then $\Pi_k^*(S_k, E_m)$ *is strictly increasing in* S_k *for any* $E_m > 0$*, satisfies* $\lim_{E_m \downarrow 0} \Pi_k^*(S_k, E_m) = 1$*, and satisfies* $\lim_{E_m \uparrow \infty} \Pi_k^*(S_k, E_m) = 0$*.*

Proof. If $E_m = 0$, then $\Pi_k(E_M, 0) = 1$ for all $E_M \geq 0$, so the unique best response is $e_i = 0$ and hence $E_M^*(S_k, 0) = 0$.

Fix $E_m > 0$ and $S_k > 0$. Under symmetry, Equation 7 is equivalent to

$$\frac{E_M}{N_M}(S_k E_M + E_m)^2 = \Lambda S_k E_m. \tag{9}$$

Define $f(E_M) \equiv \frac{E_M}{N_M}(S_k E_M + E_m)^2 - \Lambda S_k E_m$. The function f is continuous on $[0, \infty)$ with $f(0) = -\Lambda S_k E_m < 0$, and

$$f'(E_M) = \frac{1}{N_M}(S_k E_M + E_m)(3S_k E_M + E_m) > 0$$

for all $E_M \geq 0$. Moreover, $\lim_{E_M \to \infty} f(E_M) = +\infty$. Hence there exists a unique $E_M^*(S_k, E_m) > 0$ satisfying Equation 9.

To establish monotonicity in S_k, define effective enforcement $Z_k \equiv S_k E_M^*(S_k, E_m)$ and rewrite Equation 9 as

$$Z_k(Z_k + E_m)^2 = N_M \Lambda S_k^2 E_m. \tag{10}$$

For any fixed $E_m > 0$, the left-hand side is strictly increasing in Z_k, while the right-hand side is strictly increasing in S_k^2. Therefore Z_k is strictly increasing in S_k. Since $\Pi_k^*(S_k, E_m) = Z_k/(Z_k + E_m)$ is strictly increasing in Z_k, it follows that Π_k^* is strictly increasing in S_k.

Finally, let $r_k \equiv Z_k/E_m$. Dividing Equation 10 by E_m^3 yields $r_k(r_k + 1)^2 = N_M \Lambda S_k^2/E_m^2$. As $E_m \downarrow 0$, the right-hand side diverges to $+\infty$, so $r_k \to +\infty$ and hence $\Pi_k^*(S_k, E_m) = r_k/(r_k + 1) \to 1$. As $E_m \uparrow \infty$, the right-hand side converges to 0, so $r_k \to 0$ and hence $\Pi_k^*(S_k, E_m) \to 0$. ■

Combining Equation 3 and Equation 8, the equilibrium security level under incumbent party k and symbolic policy d_k is

$$\Pi_k^*(d_k; S_k) \;\equiv\; \Pi_k^*\big(S_k, E_m(d_k)\big) = \Pi_k^*\big(S_k, \gamma d_k^\sigma\big). \tag{11}$$

Two implications will be useful below. First, as $d_k \downarrow 0$, minority mobilization vanishes, equilibrium security converges to one under either party, and competence differences become electorally irrelevant. Second, when $d_k > 0$ induces a non-trivial threat, higher S_k translates into higher equilibrium security by raising effective enforcement Z_k.

2.3 Connection to Contest Theory: Manipulating the Group Contest

This subsection makes explicit the model's link to contest theory. In canonical contest environments, a (typically neutral) principal designs prizes or rules to shape equilibrium effort and performance (e.g., Barut and Kovenock (1998)). Recent work also emphasizes that increasing the intensity of competition can *discourage* effort, yielding non-monotone response patterns (e.g., Fang et al. (2020)). In contrast, my political economy environment features a *biased* designer: the incumbent party manipulates the contest *indirectly* through symbolic policy. By choosing d_k, the incumbent selects the Minority's equilibrium mobilization $E_m(d_k)$ and thereby the "threat" faced in the decentralized Majority subgame. This is the precise sense in which provocative symbolic policy can be interpreted as "arming the enemy"—an electoral strategy that increases the intensity of the group contest in order to activate Majority enforcement and magnify the return to security competence.

2.3.1 Comparative statics on asymmetric technology S_k

Fix $E_m > 0$ and consider how Majority mobilization responds to an increase in state security competence S_k. Define

$$F(E_M, S_k; E_m) \equiv \frac{E_M}{N_M} - \Lambda \frac{S_k E_m}{(S_k E_M + E_m)^2}.$$

The symmetric equilibrium $E_M^*(S_k, E_m)$ is the unique solution to $F(E_M, S_k; E_m) = 0$.

Lemma 2 (State capacity and Majority effort) *Fix $E_m > 0$. The equilibrium mapping $S_k \mapsto E_M^*(S_k, E_m)$ is continuously differentiable and satisfies*

$$\frac{\partial E_M^*(S_k, E_m)}{\partial S_k} = -\frac{\frac{\partial F}{\partial S_k}(E_M^*, S_k; E_m)}{\frac{\partial F}{\partial E_M}(E_M^*, S_k; E_m)} = -\frac{\Lambda E_m \big(S_k E_M^*(S_k, E_m) - E_m\big)}{(S_k E_M^*(S_k, E_m) + E_m)^3 \left[\frac{1}{N_M} + \frac{2\Lambda S_k^2 E_m}{(S_k E_M^*(S_k, E_m) + E_m)^3}\right]}. \tag{12}$$

In particular, $\frac{\partial E_M^*}{\partial S_k} > 0$ if and only if $S_k E_M^*(S_k, E_m) < E_m$. Moreover, effective enforcement $Z_k^*(S_k, E_m) \equiv S_k E_M^*(S_k, E_m)$ is strictly increasing in S_k.

Proof. Fix $E_m > 0$. By Lemma 1, $E_M^*(S_k, E_m)$ is uniquely defined for each $S_k > 0$. Moreover,

$$\frac{\partial F}{\partial E_M}(E_M, S_k; E_m) = \frac{1}{N_M} + \frac{2\Lambda S_k^2 E_m}{(S_k E_M + E_m)^3} > 0$$

for all $E_M \geq 0$ and $S_k > 0$, so the Implicit Function Theorem implies that $E_M^*(S_k, E_m)$ is

continuously differentiable in S_k.

Differentiating $F(E_M^*(S_k, E_m), S_k; E_m) = 0$ with respect to S_k yields

$$\frac{\partial F}{\partial E_M}\frac{\partial E_M^*}{\partial S_k} + \frac{\partial F}{\partial S_k} = 0,$$

where

$$\frac{\partial F}{\partial S_k}(E_M, S_k; E_m) = -\Lambda E_m \frac{\partial}{\partial S_k}\Big(\frac{S_k}{(S_k E_M + E_m)^2}\Big) = \Lambda E_m \frac{S_k E_M - E_m}{(S_k E_M + E_m)^3}.$$

Substituting into the total derivative expression yields Equation 12 and the sign characterization.

The final claim follows from Lemma 1, which shows that $Z_k^*(S_k, E_m)$ is strictly increasing in S_k and that $\Pi_k^*(S_k, E_m) = Z_k^*/(Z_k^* + E_m)$ is therefore increasing in S_k. ■

Corollary 1 (A technology threshold) *Fix $E_m > 0$ and define $\overline{S}(E_m) \equiv 2E_m/\sqrt{N_M\Lambda}$. Then $E_M^*(S_k, E_m)$ is strictly increasing in S_k for all $S_k \in (0, \overline{S}(E_m))$, strictly decreasing for all $S_k \in (\overline{S}(E_m), \infty)$, and satisfies $E_M^*(\overline{S}(E_m), E_m) = \sqrt{N_M\Lambda}/2$.*

Proof. By Lemma 2, $\partial E_M^*(S_k, E_m)/\partial S_k = 0$ if and only if $S_k E_M^*(S_k, E_m) = E_m$. Substituting this restriction into Equation 7 yields $E_M^*/N_M = \Lambda/(4E_M^*)$, so $E_M^* = \sqrt{N_M\Lambda}/2$ at any stationary point. Therefore any stationary point must satisfy $S_k = E_m/E_M^* = 2E_m/\sqrt{N_M\Lambda}$, proving the expression for $\overline{S}(E_m)$ and the value at the peak.

To see that the stationary point is unique and is a global maximum, define $r \equiv S_k E_M^*(S_k, E_m)/E_m$. As in the proof of Lemma 1, r satisfies $r(r+1)^2 = N_M\Lambda S_k^2/E_m^2$, which is strictly increasing in S_k. Hence r crosses one exactly once, and Lemma 2 implies that $\partial E_M^*/\partial S_k$ changes sign exactly once (from positive to negative) at $S_k = \overline{S}(E_m)$. ■

2.3.2 Reaction functions and strategic complements/substitutes

For a fixed incumbent technology S_k, Equation 7 defines an implicit reaction function $E_M^*(S_k, E_m)$ mapping the Minority's mobilization into the Majority's equilibrium enforcement. This reaction is generically non-monotone: small threats discipline free-riding and raise enforcement, whereas sufficiently intense threats can discourage enforcement and reduce E_M^*.

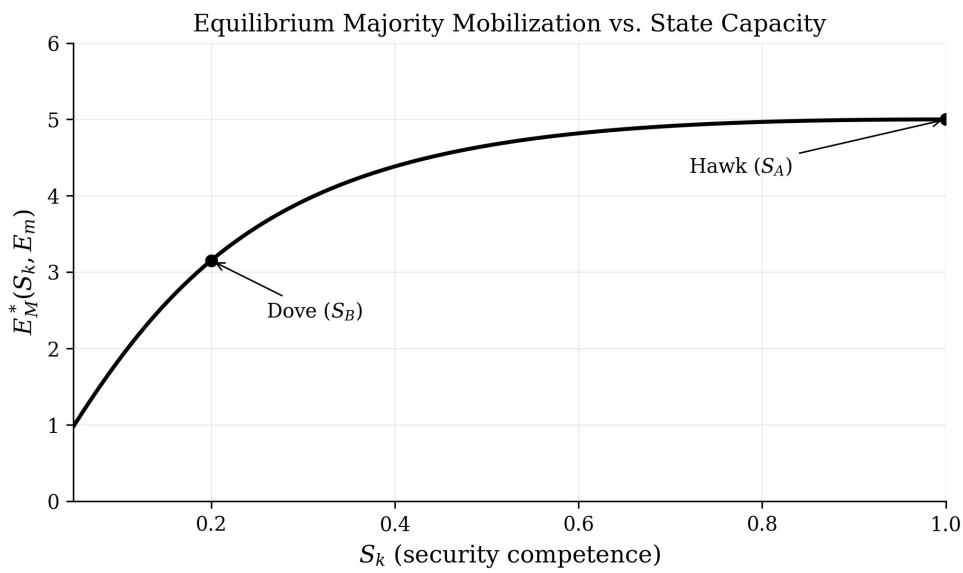

FIGURE 1 Majority mobilization as a function of asymmetric technology. The figure plots the symmetric equilibrium $E_M^*(S_k, E_m)$ implied by Equation 7 as S_k varies, holding $(N_M, \Lambda, E_m) = (100, 1, 5)$ fixed. The plotted range focuses on the increasing branch $S_k \leq \overline{S}(E_m)$ from Corollary 1, on which higher state capacity raises the marginal return to individual effort in the contest success function Equation 4 and alleviates the Majority's free-rider problem.

Lemma 3 (Inverted-U reaction and the peak threat level) *Fix $S_k > 0$. The equilibrium reaction function $E_m \mapsto E_M^*(S_k, E_m)$ is single-peaked on $(0, \infty)$, with a unique maximizer at*

$$E_m^{\text{peak}}(S_k) = \frac{S_k\sqrt{N_M\Lambda}}{2}. \tag{13}$$

At the peak, the contest is locally "balanced" in the sense that $S_k E_M^*(S_k, E_m^{\text{peak}}) = E_m^{\text{peak}}$, and the maximal Majority mobilization is $E_M^*(S_k, E_m^{\text{peak}}) = \frac{\sqrt{N_M\Lambda}}{2}$. Moreover, the reaction exhibits strategic complements for $E_m < E_m^{\text{peak}}(S_k)$ and strategic substitutes for $E_m > E_m^{\text{peak}}(S_k)$:

$$\frac{\partial E_M^*(S_k, E_m)}{\partial E_m} \begin{cases} > 0 & \text{if } E_m < E_m^{\text{peak}}(S_k), \\ = 0 & \text{if } E_m = E_m^{\text{peak}}(S_k), \\ < 0 & \text{if } E_m > E_m^{\text{peak}}(S_k). \end{cases}$$

Proof. Fix $S_k > 0$ and $E_m > 0$. By Lemma 1, $E_M^*(S_k, E_m)$ is the unique solution to $F(E_M, S_k; E_m) = 0$. Since $\frac{\partial F}{\partial E_M} > 0$ everywhere, the Implicit Function Theorem implies that

$E_M^*(S_k, E_m)$ is continuously differentiable in E_m on $(0, \infty)$. Implicit differentiation yields

$$\frac{\partial E_M^*(S_k, E_m)}{\partial E_m} = -\frac{\frac{\partial F}{\partial E_m}(E_M^*, S_k; E_m)}{\frac{\partial F}{\partial E_M}(E_M^*, S_k; E_m)} = \frac{\Lambda S_k\big(S_k E_M^*(S_k, E_m) - E_m\big)}{(S_k E_M^*(S_k, E_m) + E_m)^3\left[\frac{1}{N_M} + \frac{2\Lambda S_k^2 E_m}{(S_k E_M^*(S_k, E_m)+E_m)^3}\right]}. \tag{14}$$

Hence $\frac{\partial E_M^*}{\partial E_m} = 0$ if and only if $S_k E_M^*(S_k, E_m) = E_m$, and the sign of $\frac{\partial E_M^*}{\partial E_m}$ coincides with the sign of $S_k E_M^*(S_k, E_m) - E_m$.

To establish single-peakedness and compute the maximizer, define $r \equiv S_k E_M^*(S_k, E_m)/E_m$. As in the proof of Lemma 1, r satisfies $r(r+1)^2 = N_M \Lambda S_k^2 / E_m^2$. The right-hand side is strictly decreasing in E_m, while the left-hand side is strictly increasing in r, so r is strictly decreasing in E_m. Consequently, there exists a unique $E_m > 0$ such that $r = 1$, i.e., $S_k E_M^*(S_k, E_m) = E_m$, and by Equation 14 the reaction function increases up to that point and decreases thereafter.

At the peak, $r = 1$ implies $E_m = S_k E_M^*$. Substituting this restriction into Equation 7 yields $E_M^*/N_M = \Lambda/(4E_M^*)$, hence $E_M^* = \sqrt{N_M \Lambda}/2$ and $E_m^{\text{peak}}(S_k) = S_k\sqrt{N_M \Lambda}/2$, proving Equation 13. ■

Lemma 3 tightens the connection to contest theory in two ways. First, the inverted-U best response is a precise formulation of the "discouragement" logic: beyond the peak, additional opposition strength reduces the equilibrium return to Majority enforcement and crowds out effort, as emphasized by Fang et al. (2020). Second, the peak $E_m^{\text{peak}}(S_k)$ increases linearly in S_k: higher state capacity expands the range of threats for which Majority enforcement responds positively to opposition mobilization. This provides a contest-theoretic mechanism through which a high-S_k incumbent can alleviate the Majority's collective action problem.

2.3.3 The inefficiency of the contest: rent dissipation under biased design

Contest theory often measures the resource cost of conflict by the sum of equilibrium efforts (rent dissipation). In my environment, the incumbent's symbolic policy is the primitive "design" instrument that determines the contest intensity by shifting $E_m(d)$.

Definition 1 (Rent dissipation) *For an incumbent with technology S_k, define total rent dissipation under symbolic policy d as*

$$RD(d, S_k) \equiv E_M^*\big(S_k, E_m(d)\big) + E_m(d) = E_M^*\big(S_k, \gamma d^\sigma\big) + \gamma d^\sigma. \tag{15}$$

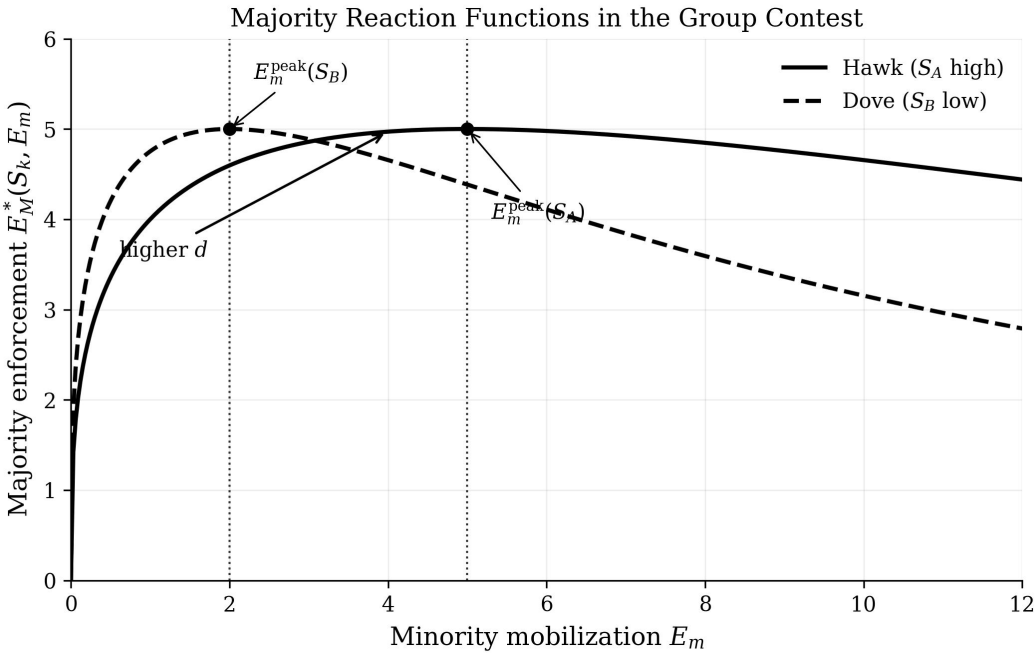

FIGURE 2 Majority reaction functions and strategic complements/substitutes. The figure plots the equilibrium response $E_M^*(S_k, E_m)$ from Equation 7 as a function of E_m, for two technology levels: a high-capacity Hawk (S_A) and a low-capacity Dove (S_B). Each curve is single-peaked at $E_m^{\text{peak}}(S_k)$ from Lemma 3. For $E_m < E_m^{\text{peak}}(S_k)$, enforcement and opposition mobilization are strategic complements ($\partial E_M^*/\partial E_m > 0$); for $E_m > E_m^{\text{peak}}(S_k)$ they are strategic substitutes ($\partial E_M^*/\partial E_m < 0$). The arrow illustrates how a more provocative symbolic policy d raises $E_m(d)$ and thereby moves the economy along the high-S_k reaction curve, activating Majority enforcement. Illustration uses $(N_M, \Lambda, S_A, S_B) = (100, 1, 1.0, 0.4)$.

Proposition 1 (Rent dissipation and biased contest design) *For any $S_k > 0$, rent dissipation satisfies $RD(0, S_k) = 0$ and $RD(d, S_k) > 0$ for all $d > 0$. Moreover, the politically optimal symbolic policy generically features strictly positive dissipation: in any equilibrium in which the incumbent chooses $d_k^* > 0$, the realized contest dissipates resources $RD(d_k^*, S_k) > 0$.*

Proof. Since $E_m(0) = 0$ by Equation 3, Lemma 1 implies $E_M^*(S_k, 0) = 0$ and hence $RD(0, S_k) = 0$. If $d > 0$, then $E_m(d) = \gamma d^\sigma > 0$, and Lemma 1 implies $E_M^*(S_k, E_m(d)) > 0$. Therefore $RD(d, S_k) > 0$ for all $d > 0$.

The final statement follows immediately: if an incumbent chooses $d_k^* > 0$ in equilibrium, then the induced dissipation is $RD(d_k^*, S_k) > 0$ by the preceding argument. In particular, combining this observation with Proposition 2 implies that strategic provocation ("arming

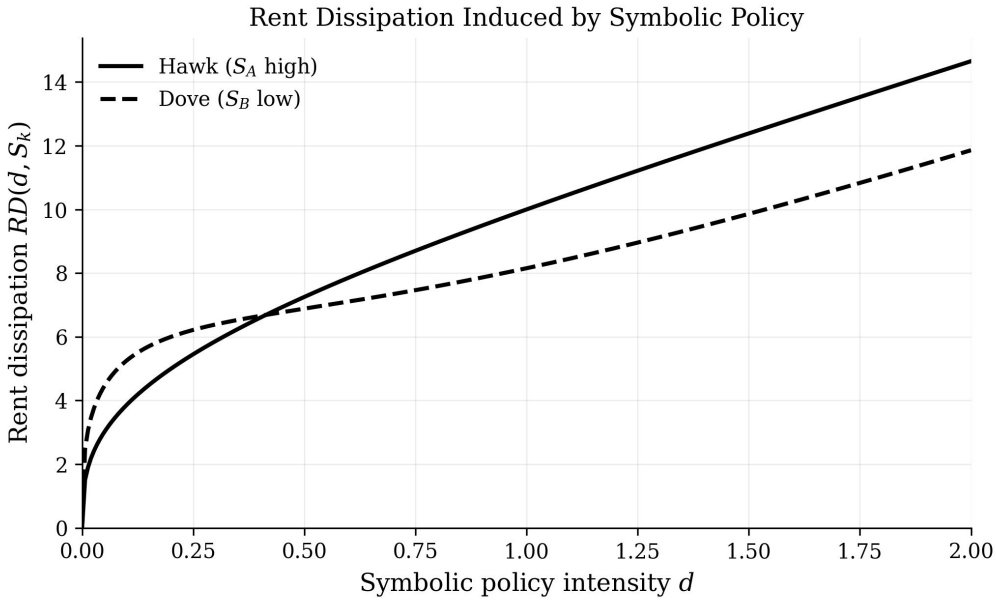

FIGURE 3 Rent dissipation under symbolic policy. The figure plots $RD(d, S_k)$ from Equation 15 against symbolic policy intensity d, for two levels of state capacity. By inducing $E_m(d) > 0$, symbolic policy creates a contest that wastes real resources through both Minority mobilization and Majority enforcement. Unlike the neutral contest designer studied in standard contest design settings (e.g., Barut and Kovenock (1998)), the political incumbent is biased: the Hawk may deliberately choose a strictly positive d—and hence strictly positive dissipation—because the induced conflict helps expand the Security Premium in the election. Illustration uses $(N_M, \Lambda, \gamma, \sigma, S_A, S_B) = (100, 1, 5, 1, 1.0, 0.2)$.

the enemy") in equilibrium is necessarily accompanied by strictly positive rent dissipation. ■

Taken together, Parts A–C formalize a tight connection between symbolic politics and contest theory: by choosing d_k, the incumbent selects the threat environment $E_m(d_k)$ and thereby manipulates the equilibrium mapping from security competence S_k into realized security $\Pi_k^*(d_k; S_k)$. This mechanism is central to the political economy analysis below.

3 Voter Preferences and Electoral Competition

Preferences in the Majority electorate are characterized by heterogeneity. Voter i derives utility from a weighted combination of physical security and economic consumption, augmented by an idiosyncratic partisan bias. The per-period utility from electing Party

$k \in \{A, B\}$ is:

$$U_i^k = \lambda \Pi_k^*(d_k; S_k) + (1-\lambda)u(g_k) - \chi(d_k) + \xi_i, \tag{16}$$

where $\lambda \in (0,1)$ is a fixed (exogenous) weight on security, and $\chi(d)$ is a convex cost of symbolic regulation borne by the Majority (economic disruption, administrative burden, or moderate-voter backlash), with $\chi'(d) > 0$ and $\chi''(d) > 0$. The term ξ_i captures an idiosyncratic preference shock in favor of the Hawk, distributed according to a cumulative distribution function $\Phi(\cdot)$ with median zero (a normalization).

A citizen i votes for Party A if $U_i^A \geq U_i^B$. Substituting (16), the indifference condition for the *swing voter* $\hat{\xi}$ is:

$$\hat{\xi} = (1-\lambda)[u(g_B) - u(g_A)] + [\chi(d_A) - \chi(d_B)] - \lambda\Big(\Pi_A^*(d_A; S_A) - \Pi_B^*(d_B; S_B)\Big). \tag{17}$$

The first term captures the Hawk's economic deficit. The second term captures the direct cost of symbolic policy for the (swing) Majority electorate. The third term is the Security Premium. Party A secures an electoral victory if the median voter prefers A, corresponding to $\Phi(-\hat{\xi}) > 1/2$.

4 Baseline Analysis

The political equilibrium is the subgame perfect equilibrium of the three-stage game described above. In the policy stage, the incumbent chooses (d_k, g_k). In the contest stage, mobilization levels $E_m(d_k)$ and $E_M^*(S_k, E_m)$ are determined as in Equation 3–Equation 11. In the election stage, voters compare indirect utilities and the swing-voter threshold $\hat{\xi}$ in Equation 17 pins down the re-election probability.

We first characterize the voting behavior of the Majority. A citizen i votes for Party A if $U_i^A \geq U_i^B$. Substituting the utility function from (16) yields the swing-voter threshold $\hat{\xi}$ in Equation 17. Party A secures an electoral victory if the median voter prefers A, which corresponds to $\Phi(-\hat{\xi}) > 1/2$. The incumbent's objective is therefore to maximize the vote share by manipulating $\hat{\xi}$.

The central mechanism of the model relies on the interaction between opposition strength and the collective action problem. Holding fixed the threat level E_m faced at the election, define the *Security Premium* as the difference in equilibrium security provided by the Hawk versus the Dove,

$$\Delta\Pi(E_m) \;\equiv\; \Pi_A^*(S_A, E_m) - \Pi_B^*(S_B, E_m), \tag{18}$$

where $\Pi_k^*(S_k, E_m)$ is defined in Equation 8.

Lemma 4 (Comparative Advantage in Conflict) *Suppose $S_A > S_B$. The Security Premium $\Delta\Pi(E_m)$ satisfies $\Delta\Pi(0) = 0$, $\Delta\Pi(E_m) > 0$ for all $E_m > 0$, and $\lim_{E_m \uparrow \infty} \Delta\Pi(E_m) = 0$. Moreover, there exists $\overline{E} > 0$ such that $\Delta\Pi(E_m)$ is strictly increasing on $[0, \overline{E}]$.*

Proof. The boundary conditions follow from Lemma 1. In particular, $\lim_{E_m \downarrow 0} \Pi_k^*(S_k, E_m) = 1$ for any $S_k > 0$, so $\Delta\Pi(0) = 0$. Moreover, $\Pi_k^*(S_k, E_m)$ is strictly increasing in S_k for any $E_m > 0$, so $\Delta\Pi(E_m) > 0$ whenever $E_m > 0$. Finally, $\lim_{E_m \uparrow \infty} \Pi_k^*(S_k, E_m) = 0$, so $\lim_{E_m \uparrow \infty} \Delta\Pi(E_m) = 0$.

To show that $\Delta\Pi$ is strictly increasing near $E_m = 0$, write equilibrium security as $\Pi_k^*(S_k, E_m) = r_k/(r_k + 1)$, where $r_k \equiv Z_k/E_m$ and Z_k satisfies Equation 10. Dividing Equation 10 by E_m^3 yields

$$r_k(r_k + 1)^2 = \frac{N_M \Lambda S_k^2}{E_m^2}. \tag{19}$$

As $E_m \downarrow 0$, the right-hand side of Equation 19 diverges to $+\infty$, so $r_k \to +\infty$. Using $r_k(r_k + 1)^2 = r_k^3(1 + o(1))$ as $r_k \to \infty$, Equation 19 implies $r_k = (N_M \Lambda)^{1/3} S_k^{2/3} E_m^{-2/3}(1 + o(1))$. Therefore

$$\Pi_k^*(S_k, E_m) = 1 - \frac{1}{r_k + 1} = 1 - (N_M \Lambda)^{-1/3} S_k^{-2/3} E_m^{2/3}(1 + o(1)).$$

It follows that, as $E_m \downarrow 0$,

$$\Delta\Pi(E_m) = (N_M \Lambda)^{-1/3} E_m^{2/3} \left(S_B^{-2/3} - S_A^{-2/3} \right)(1 + o(1)),$$

which is strictly increasing in E_m for E_m sufficiently small. Hence there exists $\overline{E} > 0$ such that $\Delta\Pi$ is strictly increasing on $[0, \overline{E}]$. ■

The Security Premium is therefore an indirect function of symbolic policy through the opposition's reaction, $\Delta\Pi(E_m(d))$. In the baseline electoral problem, the incumbent chooses symbolic regulation to trade off a larger Security Premium against the direct electoral cost $\chi(d)$.

Proposition 2 (Strategic Provocation and an Economic-Deficit Cutoff) *Assume $\chi(0) = 0$ and $d_B^* = 0$ in equilibrium. Let the Hawk's economic disadvantage be $\Delta u \equiv u(g_B) - u(g_A) > 0$. Then any political equilibrium in which Party A wins office satisfies $d_A^* > 0$ and hence $E_m(d_A^*) > 0$.*

Moreover, define the cutoff

$$\overline{\Delta u} \;\equiv\; \frac{1}{1-\lambda}\sup_{d\geq 0}\Big\{\lambda\Delta\Pi(E_m(d)) - \chi(d)\Big\}. \tag{20}$$

Party A can win the election if and only if $\Delta u \leq \overline{\Delta u}$.

Proof. Fix $d_B^* = 0$. Since Party A maximizes its vote share, it equivalently minimizes the swing-voter threshold $\hat{\xi}$ in Equation 17. Under $\chi(0) = 0$ and $d_B = 0$, the threshold is

$$\hat{\xi}(d) \;=\; (1-\lambda)\Delta u + \chi(d) - \lambda\Delta\Pi(E_m(d)). \tag{21}$$

Equivalently, Party A chooses d to maximize

$$F(d) \;\equiv\; \lambda\Delta\Pi(E_m(d)) - \chi(d). \tag{22}$$

If F is differentiable on $(0,\infty)$, any interior optimum $d_A^* > 0$ satisfies the first-order condition $\lambda\Delta\Pi'(E_m(d_A^*))E_m'(d_A^*) = \chi'(d_A^*)$. Because $E_m(0) = 0$ by Equation 3, and because in the absence of opposition mobilization ($E_m = 0$) both parties deliver full dominance (so $\Pi_A = \Pi_B = 1$), it follows that $\Delta\Pi(E_m(0)) = 0$ and hence $F(0) = 0$.

If Party A wins office in equilibrium, then the median voter weakly prefers A, i.e., $\hat{\xi}(d_A^*) \leq 0$. By Equation 21, this requires $F(d_A^*) \geq (1-\lambda)\Delta u > 0$. Since $F(0) = 0$, it follows that $d_A^* \neq 0$, so any winning equilibrium must satisfy $d_A^* > 0$ and therefore $E_m(d_A^*) > 0$.

For the cutoff, note that Party A wins if and only if there exists a feasible $d \geq 0$ such that $(1-\lambda)\Delta u \leq \lambda\Delta\Pi(E_m(d)) - \chi(d)$. Taking the supremum over d yields the bound in Equation 20. If $\Delta u \leq \overline{\Delta u}$, then there exists d achieving the inequality; if $\Delta u > \overline{\Delta u}$, then no d can satisfy it. ■

Remark. Since $\Delta u \equiv u(g_B) - u(g_A)$ is increasing in the Hawk's relative cost disadvantage in public good provision (through the induced equilibrium choice of g_A), the cutoff $\overline{\Delta u}$ in Proposition 2 can be equivalently expressed as a threshold condition on the Hawk's economic incompetence.

5 Dynamic Extension: Myopia, Forgetting, and Regime Switching

This section develops a parsimonious dynamic extension that delivers political-economy cycles—alternating high-d/low-g and low-d/high-g regimes—without allowing the preference weight on security λ to vary over time. The key ingredients are (i) aggregate shocks

that are imperfectly observed, (ii) retrospective (myopic) electoral accountability, and (iii) the regime-dependence of conflict production through S_k and the induced equilibrium mobilization $E_M^*(S_k, E_m)$.

5.1 Aggregate shocks and noisy performance

Time is discrete, $t = 0, 1, 2, \ldots$, and an election takes place each period. Let θ_t denote an aggregate economic state and ω_t an aggregate threat state. For expositional simplicity, assume each party offers stationary platforms (d_k, g_k) with $g_B > g_A$ and $d_A > d_B$.

Under a candidate party $k \in \{A, B\}$, the realized economic payoff is

$$c_{k,t} = \theta_t g_k, \tag{23}$$

and minority mobilization is

$$E_{m,t}(k) = \omega_t \gamma d_k^\sigma. \tag{24}$$

Given $(S_k, E_{m,t}(k))$, Majority mobilization is $E_{M,t}^*(S_k, E_{m,t}(k))$, and security is $\Pi_{k,t} = \Pi_k(E_{M,t}^*, E_{m,t}(k))$ as in Equation 4.

Voters do not observe (θ_t, ω_t) nor the counterfactual outcomes under the challenger. Instead, they observe noisy performance signals of consumption and disorder under the incumbent regime k_t:

$$\tilde{c}_t = c_{k_t,t} + \varepsilon_t^c, \qquad \tilde{v}_t = \left(1 - \Pi_{k_t,t}\right) + \varepsilon_t^v, \tag{25}$$

where $\tilde{v}_t$ is a reduced-form measure of observable street disorder/violence. Importantly, because $\Pi_{k,t}$ depends on S_k and the endogenous response $E_{M,t}^*(S_k, E_m)$, the same underlying threat ω_t can generate very different observed disorder across regimes.

5.2 Forward-looking voting and the conflict trap

Depending on voter patience, the dynamic environment admits two qualitatively distinct equilibrium paths. When voters are forward-looking, regime choice is governed by the discounted value of future security capacity.

Proposition 3 (Conflict Trap) *Assume voters are forward-looking with discount factor $\delta \in [0, 1)$ and form rational expectations about the future evolution of (θ_t, ω_t) under each regime. There exists a critical patience threshold $\bar{\delta} \in (0, 1)$ such that for all $\delta > \bar{\delta}$, the Hawk regime is an absorbing state in a stationary political equilibrium. In particular, conditional on a Hawk incumbent, the electorate*

retains the Hawk even after the most adverse economic realizations $\theta_t = \underline{\theta}$ in order to avoid the discrete reduction in security capacity associated with switching from S_A to S_B.

Proof. Fix an incumbent Hawk regime and suppress the idiosyncratic taste term ξ_i by considering the median voter (normalized to have $\xi_i = 0$). Let

$$\Omega(\theta, \omega) \;\equiv\; \lambda\Big(\Pi_A(\theta,\omega) - \Pi_B(\theta,\omega)\Big) - (1-\lambda)\Big(u(\theta g_B) - u(\theta g_A)\Big) - \big[\chi(d_A) - \chi(d_B)\big] \quad (26)$$

denote the one-period net advantage of the Hawk relative to the Dove in state (θ, ω). Consider the worst economic realization $\underline{\theta}$ and assume (as in the proposition) that it constitutes the most unfavorable state for the Hawk's re-election incentive.

To isolate the role of patience, consider a transitory recession at date t with $\theta_t = \underline{\theta}$, followed by reversion from $t+1$ onward to a stationary environment. In that continuation environment, the discrete gap in security capacity ($S_A > S_B$) implies that the expected per-period net advantage of the Hawk is strictly positive:

$$\overline{\Omega} \;\equiv\; \mathbb{E}\big[\Omega(\theta,\omega) \mid t+1 \text{ onward}\big] \;>\; 0. \quad (27)$$

Let $\underline{\Omega} \equiv \mathbb{E}\big[\Omega(\underline{\theta}, \omega_t)\big]$ denote the expected one-period net advantage during the recession. The forward-looking value difference between retaining the Hawk and switching regimes at date t can then be expressed (via the Bellman equations under stationarity from $t+1$ onward) as

$$\Delta V_t \;=\; \underline{\Omega} + \sum_{\tau=t+1}^{\infty} \delta^{\tau-t}\overline{\Omega} \;=\; \underline{\Omega} + \frac{\delta}{1-\delta}\overline{\Omega}. \quad (28)$$

The function $\Delta V_t(\delta)$ is continuous and strictly increasing on $[0,1)$. Moreover, $\lim_{\delta\uparrow 1} \Delta V_t(\delta) = +\infty$ by Equation 27. Hence there exists a cutoff $\bar{\delta} \in (0,1)$ such that $\Delta V_t(\delta) \geq 0$ for all $\delta \geq \bar{\delta}$. For $\delta > \bar{\delta}$ the median voter weakly prefers to retain the Hawk even when $\theta_t = \underline{\theta}$. Since $\underline{\theta}$ is the most adverse economic realization, the Hawk is retained in every state, implying that the Hawk regime is absorbing in a stationary equilibrium. ■

Intuition. High patience raises the weight placed on the continuation value of security capacity. Once a Hawk incumbent is in power, the expected discounted loss in future security from switching regimes can dominate any short-run gains in public good provision. As a result, regime cycling disappears for sufficiently large δ, and the economy converges to persistent Hawk dominance (a conflict trap).

5.3 Myopic voting and regime transitions

Let per-period utility be given by (16). A forward-looking voter with discount factor $\delta \in [0, 1)$ evaluates parties by lifetime value. The special case $\delta \to 0$ corresponds to purely retrospective (myopic) voting: the electorate selects the party that maximizes perceived one-period utility based on the current performance signals $(\tilde{c}_t, \tilde{v}_t)$.

To connect the dynamic environment to the static swing-voter logic, define the perceived one-period Hawk-selection condition

$$\lambda\Big(\mathbb{E}_t[\Pi_{A,t}] - \mathbb{E}_t[\Pi_{B,t}]\Big) \;\geq\; (1-\lambda)\Big(\mathbb{E}_t[u(\theta_t g_B)] - \mathbb{E}_t[u(\theta_t g_A)]\Big) + \big[\chi(d_A) - \chi(d_B)\big], \tag{29}$$

which is the dynamic analogue of Equation 17.

Proposition 4 (Political Cycles) *Assume voters are myopic ($\delta = 0$), and (θ_t, ω_t) are stochastic with full support on $[\underline{\theta}, \overline{\theta}] \times [\underline{\omega}, \overline{\omega}]$. Suppose (i) the posterior expectations entering Equation 29 are monotone in the underlying states,*[2] *(ii) for $g_B > g_A$, the function $\theta \mapsto u(\theta g_B) - u(\theta g_A)$ is strictly decreasing on $[\underline{\theta}, \overline{\theta}]$, and (iii) the Security Premium is increasing in minority mobilization over the policy-relevant domain (Lemma 4). Assume further that the myopic election rule selects different parties in different states (i.e., there exist (θ, ω) and (θ', ω') in the support such that Equation 29 holds in one and fails in the other). Then the political equilibrium exhibits recurrent regime switching. In particular:*

(i) (Hawk $\to$ Dove) *Conditional on a Hawk incumbent, the probability of a regime change is strictly decreasing in θ_t.*

(ii) (Dove $\to$ Hawk) *Conditional on a Dove incumbent, the probability of a regime change is strictly increasing in the threat environment ω_t, the minority's structural mobilization potential γ, and the security weight λ.*

Proof. Under myopic voting ($\delta = 0$), the median voter elects the Hawk at date t if and only if Equation 29 holds. Define the net advantage

$$\Omega_t \;\equiv\; \lambda\Big(\mathbb{E}_t[\Pi_{A,t}] - \mathbb{E}_t[\Pi_{B,t}]\Big) - (1-\lambda)\Big(\mathbb{E}_t[u(\theta_t g_B)] - \mathbb{E}_t[u(\theta_t g_A)]\Big) - \big[\chi(d_A) - \chi(d_B)\big], \tag{30}$$

so the Hawk is elected if and only if $\Omega_t \geq 0$.

2. For instance, a natural sufficient condition is that the signal structure $(\tilde{c}_t, \tilde{v}_t)$ satisfies a monotone likelihood ratio property so that $\mathbb{E}_t[\theta_t]$ and $\mathbb{E}_t[\omega_t]$ are monotone in the realized signals.

For part (i), note that for $g_B > g_A$,

$$\frac{\partial}{\partial\theta}\Big(u(\theta g_B) - u(\theta g_A)\Big) = g_B u'(\theta g_B) - g_A u'(\theta g_A), \tag{31}$$

so assumption (ii) is equivalent to $g_B u'(\theta g_B) - g_A u'(\theta g_A) < 0$ on $[\underline{\theta}, \overline{\theta}]$.[3] Combining this with assumption (i), a lower realization of θ_t strictly increases the perceived economic advantage of the Dove and therefore strictly decreases Ω_t.

Moreover, Ω_t is continuous in (θ_t, ω_t) (given continuity of u and monotone posteriors), and by hypothesis there exist states in which $\Omega_t \geq 0$ and states in which $\Omega_t < 0$. Fix any $\omega \in [\underline{\omega}, \overline{\omega}]$ and hold $\omega_t = \omega$. The previous argument implies that Ω_t is strictly increasing in θ_t on $[\underline{\theta}, \overline{\theta}]$. Hence the intermediate value theorem yields a cutoff $\theta^*(\omega) \in (\underline{\theta}, \overline{\theta})$ such that the myopic electorate selects the Hawk if and only if $\theta_t \geq \theta^*(\omega_t)$. Therefore, conditional on a Hawk incumbent, the Hawk $\to$ Dove transition event $\{\Omega_t < 0\}$ is equivalent to $\{\theta_t < \theta^*(\omega_t)\}$, which becomes (weakly) more likely under more adverse economic realizations.

For part (ii), note that minority mobilization satisfies $E_{m,t}(k) = \omega_t \gamma d_k^\sigma$, so higher ω_t or higher γ increases the threat faced under either candidate platform. By Lemma 4 (over the relevant domain), the perceived Security Premium $\mathbb{E}_t[\Pi_{A,t} - \Pi_{B,t}]$ is increasing in the threat level, so Ω_t is increasing in ω_t and γ. Moreover,

$$\frac{\partial \Omega_t}{\partial \lambda} = \Big(\mathbb{E}_t[\Pi_{A,t}] - \mathbb{E}_t[\Pi_{B,t}]\Big) + \Big(\mathbb{E}_t[u(\theta_t g_B)] - \mathbb{E}_t[u(\theta_t g_A)]\Big) > 0, \tag{32}$$

so Ω_t is increasing in λ. Therefore the probability that Ω_t crosses zero in favor of the Hawk (a Dove $\to$ Hawk transition) is increasing in ω_t, γ, and λ.

Finally, since the state process has full support and (by assumption) there exist states in which $\Omega_t \geq 0$ and states in which $\Omega_t < 0$, both transitions occur with positive probability along the realized path, yielding recurrent regime switching. ■

Intuition. Under a Hawk incumbent, high security competence S_A can translate into low realized disorder $\tilde{v}_t$ even when symbolic policy is provocative. Under myopic retrospective voting, low $\tilde{v}_t$ depresses beliefs about the latent threat and compresses the perceived Security Premium in Equation 29. A sufficiently adverse economic realization then makes the economic term dominate, triggering a Hawk $\to$ Dove transition. Under a Dove incumbent, the low competence parameter S_B undermines Majority mobilization, so unfavorable security realizations produce large drops in $\Pi_{B,t}$ and visible spikes in disorder; this widens the

3. A common sufficient condition is CRRA utility $u(c) = c^{1-\rho}/(1-\rho)$ with $\rho > 1$, in which case $g_B u'(\theta g_B) - g_A u'(\theta g_A) = \theta^{-\rho}\big(g_B^{1-\rho} - g_A^{1-\rho}\big) < 0$ for all $\theta > 0$.

perceived Security Premium and triggers a Dove → Hawk transition.

Corollary 2 (The Paradox of Competence) *Under myopic voting ($\delta = 0$), a first-order stochastic decrease in observed disorder under a Hawk incumbent (lower $\tilde{v}_t$) weakly decreases the perceived Security Premium in Equation 29 and thereby weakly increases the probability of a Hawk defeat. Consequently, conditional on a Hawk incumbent, the hazard of regime change is decreasing in observed disorder.*

Proof. Under myopic voting, the Hawk is re-elected at date t if and only if Equation 29 holds, equivalently if $\Omega_t \geq 0$ where Ω_t is defined in the proof of Proposition 4. By assumption (i) in Proposition 4 (and the monotone-likelihood-ratio sufficient condition in its footnote), a lower realization of the disorder signal $\tilde{v}_t$ weakly shifts posterior beliefs toward a lower threat environment and therefore weakly decreases the perceived Security Premium $\mathbb{E}_t[\Pi_{A,t} - \Pi_{B,t}]$.

Since Ω_t is increasing in $\mathbb{E}_t[\Pi_{A,t} - \Pi_{B,t}]$, a first-order stochastic decrease in $\tilde{v}_t$ induces a first-order stochastic decrease in Ω_t and hence weakly decreases $\mathbb{P}(\Omega_t \geq 0)$, the re-election probability of the Hawk. Equivalently, the probability of a regime change, $\mathbb{P}(\Omega_t < 0)$, weakly increases when observed disorder is stochastically lower. ■

Together, Propositions 3 and 4 characterize how voter patience selects between a conflict trap and political cycles.

6 Simulations

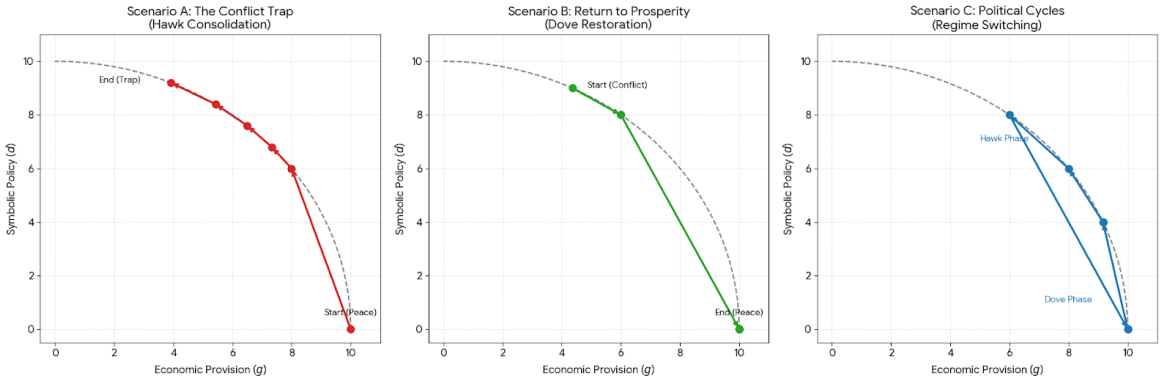

FIGURE 4 Representative simulated path of regime switching (placeholder).

This section illustrates the regime-switching logic in the dynamic extension. A natural simulation design is to specify Markov processes for (θ_t, ω_t), generate noisy performance

signals $(\tilde{c}_t, \tilde{v}_t)$, and simulate repeated elections under alternative voter discount factors δ. The key objects to track are the incumbent sequence (k_t), symbolic policy (d_{k_t}), public good provision (g_{k_t}), and the endogenous conflict variables $(E_{m,t}, E^*_{M,t}, \Pi_{k_t,t})$.

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